

2025-2026 学年第二学期《高等数学 (A 类)II》期末考试

试题 (A 卷)

(评分标准)

1. (10 分) 设 $z = z(x, y)$ 由方程 $e^{x+2y+3z} + xy = 1$ 确定, $\vec{l} = (1, 2)$. 计算 $dz|_{(0,0)}$ 和 $\frac{\partial z}{\partial \vec{l}}|_{(0,0)}$.

【解】 令

$$F(x, y, z) = e^{x+2y+3z} + xy - 1 = 0,$$

则

$$\frac{\partial F}{\partial x} = e^{x+2y+3z} + y, \quad \frac{\partial F}{\partial y} = 2e^{x+2y+3z} + x, \quad \frac{\partial F}{\partial z} = 3e^{x+2y+3z}. \quad (3')$$

由隐函数求导公式,

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} = -\frac{e^{x+2y+3z} + y}{3e^{x+2y+3z}}, \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = -\frac{2e^{x+2y+3z} + x}{3e^{x+2y+3z}}. \quad (5')$$

另一方面, 当 $x = y = 0$ 时, $z(0, 0) = 0$, 因此

$$dz = \frac{\partial z}{\partial x}\bigg|_{(0,0)} dx + \frac{\partial z}{\partial y}\bigg|_{(0,0)} dy,$$

即

$$dz|_{(0,0)} = -\frac{1}{3}dx - \frac{2}{3}dy. \quad (8')$$

方向导数为

$$\frac{\partial z}{\partial \vec{l}}\bigg|_{(0,0)} = (\nabla z)(0, 0) \cdot \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) = -\frac{\sqrt{5}}{3}. \quad (10')$$

2. (12 分) 已知函数 $u = yf\left(\frac{x}{y}\right) + xg\left(\frac{y}{x}\right)$, 其中 f, g 具有二阶连续导数, 求 $x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y}$ 的值.

【解】

$$\frac{\partial u}{\partial x} = f'\left(\frac{x}{y}\right) + g\left(\frac{y}{x}\right) - \frac{y}{x}g'\left(\frac{y}{x}\right), \quad (3')$$

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \frac{1}{y}f''\left(\frac{x}{y}\right) - \frac{y}{x^2}g'\left(\frac{y}{x}\right) + \frac{y}{x^2}g'\left(\frac{y}{x}\right) + \frac{y^2}{x^3}g''\left(\frac{y}{x}\right) \\ &= \frac{1}{y}f''\left(\frac{x}{y}\right) + \frac{y^2}{x^3}g''\left(\frac{y}{x}\right), \end{aligned} \quad (6')$$

$$\begin{aligned}\frac{\partial^2 u}{\partial x \partial y} &= -\frac{x}{y^2} f''\left(\frac{x}{y}\right) + \frac{1}{x} g'\left(\frac{y}{x}\right) - \frac{1}{x} g'\left(\frac{y}{x}\right) - \frac{y}{x^2} g''\left(\frac{y}{x}\right) \\ &= -\frac{x}{y^2} f''\left(\frac{x}{y}\right) - \frac{y}{x^2} g''\left(\frac{y}{x}\right).\end{aligned}\quad (9')$$

故

$$x \frac{\partial^2 u}{\partial x^2} + y \frac{\partial^2 u}{\partial x \partial y} = 0. \quad (12')$$

3. (10 分) 已知函数 $z = f(x, y)$ 的全微分为 $dz = 2x dx - 2y dy$, 且 $f(1, 1) = 2$. 求 $z = f(x, y)$ 在椭圆域 $D = \{(x, y) | x^2 + \frac{y^2}{4} \leq 1\}$ 上的最大值和最小值.

【解】 由 $dz = 2x dx - 2y dy$ 且 $f(1, 1) = 2$, 积分可得 $f(x, y) = x^2 - y^2 + 2$. 求一阶偏导数并令其为 0:

$$\frac{\partial f}{\partial x} = 2x = 0, \quad \frac{\partial f}{\partial y} = -2y = 0$$

解得驻点 $(0, 0)$, 函数值 $f(0, 0) = 2$. (3')

接下来考虑椭圆边界 $x^2 + \frac{y^2}{4} = 1$ 上的极值, 构造拉格朗日函数:

$$F(x, y, \lambda) = f(x, y) + \lambda \left(x^2 + \frac{y^2}{4} - 1 \right)$$

$$\begin{cases} F'_x = \frac{\partial f}{\partial x} + 2\lambda x = 2(1 + \lambda)x = 0, \\ F'_y = \frac{\partial f}{\partial y} + \frac{\lambda y}{2} = -2y + \frac{1}{2}\lambda y = 0, \\ F'_\lambda = x^2 + \frac{y^2}{4} - 1 = 0, \end{cases}$$

解得到四组临界点: (6')

$$\begin{cases} x = -1 \\ y = 0 \end{cases}, \quad \begin{cases} x = 1 \\ y = 0 \end{cases}, \quad \begin{cases} x = 0 \\ y = -2 \end{cases}, \quad \begin{cases} x = 0 \\ y = 2 \end{cases}$$

分别代入 $f(x, y)$:

$$f(\pm 1, 0) = 3, \quad f(0, \pm 2) = -2 \quad (9')$$

综上, 函数 $f(x, y)$ 在椭圆域 $D = \{(x, y) | x^2 + \frac{y^2}{4} \leq 1\}$ 上最大值为 3, 最小值为 -2 . (10')

4. (10 分) 计算二重积分 $\iint_D xy dx dy$. 其中积分区域由曲线 $y = -x$, $y = -2x$, $xy = -1$, $xy = -3$ 围成的在第二象限的闭集.

【解】作变量代换： $u = xy, v = \frac{y}{x}$. 区域 D 在变换下对应于 $D' = \{(u, v) | -3 \leq u \leq -1, -2 \leq v \leq -1\}$. 雅可比行列式：

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix} = \frac{y}{x} + \frac{y}{x} = 2v$$

$$(3') \text{ 故 } \left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \frac{1}{|2v|} = -\frac{1}{2v}.$$

于是：

$$\iint_D xy \, dx dy = \int_{-3}^{-1} u \, du \int_{-2}^{-1} \left(-\frac{1}{2v}\right) dv = -2 \ln 2$$

(10')

5. (10 分) 计算由曲面 $z^2 = \frac{x^2}{4} + \frac{y^2}{9}$ 和 $2z = \frac{x^2}{4} + \frac{y^2}{9}$ 所围成的几何体的体积.

【解】两曲面所围成的几何体在 xy 平面上的投影的边界曲线为：

$$\sqrt{\frac{x^2}{4} + \frac{y^2}{9}} = \frac{1}{2} \left(\frac{x^2}{4} + \frac{y^2}{9} \right) \iff \sqrt{\frac{x^2}{4} + \frac{y^2}{9}} = 2 \iff \frac{x^2}{16} + \frac{y^2}{36} = 1$$

即所围成几何体在 xy 平面上的投影区域为 $D_{xy} = \left\{ (x, y) \left| \frac{x^2}{16} + \frac{y^2}{36} \leq 1 \right. \right\}$.

(4')

所以所求的几何体体积为：

$$V = \iint_{D_{xy}} \left[\sqrt{\frac{x^2}{4} + \frac{y^2}{9}} - \frac{1}{2} \left(\frac{x^2}{4} + \frac{y^2}{9} \right) \right] dx dy$$

取广义极坐标变换 $x = 4r \cos \theta, y = 6r \sin \theta$, 其中 $\theta \in [0, 2\pi], r \in [0, 1]$, $|J(r, \theta)| = 24r$.

(7')

得：

$$V = 24 \int_0^{2\pi} d\theta \int_0^1 (2r - 2r^2)r \, dr = 48\pi \left(\frac{2}{3} - \frac{1}{2} \right) = 8\pi$$

(10')

6. (10 分) 计算 $\int_{\Gamma} \frac{1}{x^2 + y^2 + z^2} ds$. 其中 Γ 为曲线 $x = e^t \cos t, y = e^t \sin t, z = e^t$ 上相应于从 0 变到 2 的这段弧.

【解】

$$ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \sqrt{(e^t \cos t - e^t \sin t)^2 + (e^t \sin t + e^t \cos t)^2 + e^{2t}} dt = \sqrt{3}e^t dt$$

(5')

$$\int_{\Gamma} \frac{1}{x^2 + y^2 + z^2} ds = \int_0^2 \frac{1}{e^{2t} \cos^2 t + e^{2t} \sin^2 t + e^{2t}} \sqrt{3}e^t dt$$

$$= \int_0^2 \frac{\sqrt{3}}{2} e^{-t} dt = \left[-\frac{\sqrt{3}}{2} e^{-t} \right]_0^2 = \frac{\sqrt{3}}{2} (1 - e^{-2}) \quad (10')$$

7. (8 分) 计算 $\iint_{\Sigma} \left(\frac{x^3}{a^2} + y^3 z^3 \right) dydz + \left(\frac{y^3}{b^2} + z^3 x^3 \right) dzdx + \left(\frac{z^3}{c^2} + x^3 y^3 \right) dxdy$, 其中 Σ 为曲面 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, y \geq 0$ 的右侧.

【解】 补平面 $\Sigma_0: y = 0, \frac{x^2}{a^2} + \frac{z^2}{c^2} \leq 1$, 取左侧, 则 $\Sigma + \Sigma_0$ 构成闭合曲面, 围成区域 $\Omega: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1, y \geq 0$. 设 $P = \frac{x^3}{a^2} + y^3 z^3, Q = \frac{y^3}{b^2} + z^3 x^3, R = \frac{z^3}{c^2} + x^3 y^3$. 于是 $\frac{\partial P}{\partial x} = \frac{3x^2}{a^2}, \frac{\partial Q}{\partial y} = \frac{3y^2}{b^2}, \frac{\partial R}{\partial z} = \frac{3z^2}{c^2}$.
由高斯公式:

$$I + \iint_{\Sigma_0} = \iiint_{\Omega} 3 \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dV \quad (2')$$

在 Σ_0 上, $y = 0$, 故 $dydz = 0, dxdy = 0$ 且被积函数 $x^3 z^3$ 为奇函数, 积分区域关于 x, z 轴对称, 因此:

$$\iint_{\Sigma_0} \left(\frac{x^3}{a^2} + y^3 z^3 \right) dydz + \left(\frac{y^3}{b^2} + z^3 x^3 \right) dzdx + \left(\frac{z^3}{c^2} + x^3 y^3 \right) dxdy = 0$$

原式化简为:

$$I = 3 \iiint_{\Omega} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dV \quad (5')$$

作变换: $x = ar \sin \varphi \cos \theta, y = br \sin \varphi \sin \theta, z = cr \cos \varphi$. 雅可比行列式 $J = abcr^2 \sin \varphi$, 积分限: $0 \leq r \leq 1, 0 \leq \varphi \leq \pi, 0 \leq \theta \leq \pi$. 且 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = r^2$.

$$\begin{aligned} \iiint_{\Omega} \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dV &= \int_0^\pi d\theta \int_0^\pi d\varphi \int_0^1 r^2 \cdot abcr^2 \sin \varphi dr \\ &= abc \int_0^\pi d\theta \int_0^\pi \sin \varphi d\varphi \int_0^1 r^4 dr = abc \cdot \pi \cdot 2 \cdot \frac{1}{5} = \frac{2\pi abc}{5} \end{aligned}$$

所以, $I = 3 \cdot \frac{2\pi abc}{5} = \frac{6\pi abc}{5}$. (8')

8. (12 分) 设 L 为闭区域 $D = \{(x, y) : 0 \leq x \leq 1, x^3 \leq y \leq \sqrt[3]{x}\}$ 的边界, 取逆时针方向, 计算曲线积分 $I = \int_L (x^3 y + \cos x + e^y + \ln(e^x + e^y)) dx + (2x - 2 \sin y + xy^3 + xe^y) dy$.

【解】 显然交点为 $(0, 0)$ 和 $(1, 1)$. 令 $P = x^3 y + \cos x + e^y + \ln(e^x + e^y), Q = 2x - 2 \sin y + xy^3 + xe^y$, 则:

$$\frac{\partial P}{\partial y} = x^3 + e^y + \frac{e^y}{e^x + e^y}, \quad \frac{\partial Q}{\partial x} = 2 + y^3 + e^y$$

已知边界取逆时针, 故利用格林公式可知:

$$\begin{aligned} I &= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \iint_D \left(2 + y^3 - x^3 - \frac{e^y}{e^x + e^y} \right) dx dy \\ &= 2 \iint_D dx dy + \iint_D (y^3 - x^3) dx dy - \iint_D \frac{e^y}{e^x + e^y} dx dy \end{aligned} \quad (3')$$

(1) 对于 $\iint_D dx dy$: 其中 $D = \{(x, y) : 0 \leq x \leq 1, x^3 \leq y \leq \sqrt[3]{x}\}$ 故

$$\iint_D dx dy = \int_0^1 dx \int_{x^3}^{\sqrt[3]{x}} dy = \int_0^1 (\sqrt[3]{x} - x^3) dx = \left(\frac{3}{4} x^{\frac{4}{3}} - \frac{1}{4} x^4 \right) \Big|_0^1 = \frac{1}{2} \quad (6')$$

(2) 对于 $\iint_D (y^3 - x^3) dx dy$: 因为积分区域关于 $y = x$ 对称, 所以可以利用轮换对称性可知:
 $\iint_D (y^3 - x^3) dx dy = \iint_D (x^3 - y^3) dx dy$. 那么:

$$\iint_D (y^3 - x^3) dx dy = \frac{1}{2} \left[\iint_D (x^3 - y^3) dx dy + \iint_D (y^3 - x^3) dx dy \right] = \frac{1}{2} \iint_D 0 dx dy = 0 \quad (8')$$

(3) 对于 $\iint_D \frac{e^y}{e^x + e^y} dx dy$: 因为积分区域 D 关于 $y = x$ 对称, 所以可以利用轮换对称性可知:

$$\iint_D \frac{e^y}{e^x + e^y} dx dy = \iint_D \frac{e^x}{e^y + e^x} dx dy \quad (11')$$

so:

$$\iint_D \frac{e^y}{e^x + e^y} dx dy = \frac{1}{2} \left[\iint_D \frac{e^y}{e^x + e^y} dx dy + \iint_D \frac{e^x}{e^y + e^x} dx dy \right] = \frac{1}{2} \iint_D dx dy = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

综合 (1)(2)(3) 可知: $I = 2 \times \frac{1}{2} + 0 - \frac{1}{4} = \frac{3}{4}$. (12')

9. (每小题 3 分, 共 12 分) 判断下列级数和反常积分的敛散性, 并给出分析过程.

1. $\sum_{n=1}^{\infty} \left(\frac{n}{n+1} \right)^{n^2}$

2. $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{\ln^{2026}(n+1)}{n}$

3. $\int_1^{+\infty} \frac{dx}{x\sqrt{1+x+x^2}}$

4. $\int_0^1 \frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} dx$

【解】

(1) 因为 $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n} \right)^n} = \frac{1}{e} < 1$, 所以级数收敛. (3')

(2) 设 $a_n = \frac{\ln^{2026}(n+1)}{n}$, $f(x) = \frac{\ln^{2026}(x+1)}{x}$.

$$\lim_{x \rightarrow +\infty} \frac{\ln^{2026}(x+1)}{x} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow +\infty} \frac{2026 \cdot \ln^{2025}(x+1) \cdot \frac{1}{x+1}}{1} = \lim_{x \rightarrow +\infty} \frac{2026 \ln^{2025}(x+1)}{x+1} = \dots = 0 \quad (2')$$

又 $f'(x) = \frac{\ln^{2025}(x+1)}{x^2} \left(\frac{2026x}{x+1} - \ln(x+1) \right) < 0$. 故 $a_n > 0$, 单调递减且趋于 0, 由 Leibniz 判别法, 知级数收敛. (3')

(3) 因为 $\lim_{x \rightarrow +\infty} x^2 \cdot \frac{1}{x\sqrt{1+x+x^2}} = \lim_{x \rightarrow +\infty} \frac{x}{\sqrt{1+x+x^2}} = 1$, 且 $(p = 2 > 1)$. 故无穷积分 $\int_1^{+\infty} \frac{dx}{x\sqrt{1+x+x^2}}$ 收敛. (3')

(4) 瑕点为 $x = 0$ 和 $x = 1$:

$$\int_0^1 \frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} dx = \int_0^{1/2} \frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} dx + \int_{1/2}^1 \frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} dx$$

因为当 $x \rightarrow 0^+$ 时, $\frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} \sim \frac{1}{\sqrt{x}}$, 且 $\int_0^{1/2} \frac{1}{\sqrt{x}} dx$ 收敛, 故前段积分收敛. (2')

又因为当 $x \rightarrow 1^-$ 时, $\frac{e^x}{\sqrt[3]{1-x}\sin(\sqrt{x})} \sim C \cdot \frac{1}{(1-x)^{1/3}}$, 而 $\int_{1/2}^1 \frac{dx}{(1-x)^{1/3}}$ 收敛, 故后段积分也收敛. 故原反常积分收敛 (3')

10. (6 分) 设 $a_0 = 1, a_1 = -2, a_2 = \frac{7}{2}, a_{n+1} = -\left(1 + \frac{1}{n+1}\right) a_n$ ($n \geq 2$). 证明当 $|x| < 1$ 时幂级数 $\sum_{n=0}^{\infty} a_n x^n$ 收敛, 并求其和函数 $S(x)$.

【解】 由 $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim_{n \rightarrow \infty} \frac{n+1}{n+2} = 1$. 得级数的收敛半径 $R = 1$, 故当 $|x| < 1$ 时幂级数 $\sum_{n=0}^{\infty} a_n x^n$ 收敛. (2')

由已知可知, 当 $n \geq 3$ 时, 有:

$$a_n = -\frac{n+1}{n} a_{n-1} = -\frac{n+1}{n} \left(-\frac{n}{n-1} \right) a_{n-2} = \dots = (-1)^{n-2} \frac{n+1}{3} a_2 = (-1)^{n-2} \frac{7}{6} (n+1)$$

因此和函数:

$$S(x) = a_0 + a_1 x + a_2 x^2 + \sum_{n=3}^{\infty} a_n x^n = 1 - 2x + \frac{7}{2} x^2 + \sum_{n=3}^{\infty} (-1)^n \frac{7}{6} (n+1) x^n \quad (4')$$

令 $g(x) = \sum_{n=3}^{\infty} (-1)^n (n+1) x^n$. 由于:

$$\int_0^x g(t) dt = \sum_{n=3}^{\infty} \int_0^x (-1)^n (n+1) t^n dt = - \sum_{n=3}^{\infty} (-x)^{n+1} = \frac{-x^4}{1+x}$$

两边求导, 得 $g(x) = -\left(\frac{x^4}{1+x}\right)' = -\frac{4x^3+3x^4}{(1+x)^2}$.

故 $S(x) = \frac{1}{(1+x)^2} \left(1 + \frac{x^2}{2} + \frac{x^3}{3}\right), \quad x \in (-1, 1).$ (6')